a

scRNA-seq

Expression state

One dot = one clone

One dot = one cell

Clonal programme 1

Clonal programme 2

Clonal programme 3

scBCR-seq

Heavy + light chains

H V D J

junction

L V J

B cell

Paired by cell barcode

Clones within each donor

Threadfin

Clone embeddings

Fréchet mean

in feature space

Clone embedding

b

Threadfin integrates scRNA-seq and BCR data into clone embeddings to infer relationships between clones

Algorithm pipeline

1 scRNA-seq

scBCR-seq

2 Fréchet mean

3 clone embedding

scRNA UMAP

Clonal coherence test

4

shuffled

500 permutations within each sample

observed

P < 0.01

0 0.2 0.4

clonal ICC = $\tau^2 / (\tau^2 + \sigma^2)$

Biological inference

5

scRNA UMAP

clone embedding

Threadfin

6

antigen binding

fate mapping

position along clone embedding